

Vibha Masti

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EDUCATION

Carnegie Mellon University

Master of Science, Language Technology Institute (MCDS)

Pittsburgh, PA

Dec 2024

GPA: 4.15/4.00. Courses: DL Systems (PhD), On-Device ML (PhD), Intro to DL (PhD), Cloud Computing (MS), Intro to ML (MS)

PES University

Bachelor of Technology, Computer Science and Engineering

Bangalore, India

May 2023

GPA: 9.84/10.00. Courses: Natural Language Processing, Machine Intelligence, Data Analytics, Big Data

Awards: Prof. CNR Rao Scholarship (top 2%), Immense Contribution Award

EXPERIENCE

Microsoft AI

Data & Applied Scientist

Redmond, WA

Mar 2025 – Present

- Distributed training, large-scale ETL, daily retraining, feature engineering. Conversion modeling with counterfactual objectives.

Microsoft AI

Data & Applied Scientist Intern

Redmond, WA

May 2024 – Aug 2024

- Scalable model-agnostic hyperparameter optimization and neural architecture search. Deployed lightweight NNs achieving 99% baseline performance with 1.67% parameter count.

Apple

Machine Learning Intern

Bangalore, India

Jan 2023 – Jun 2023

- Created a few-shot multi-label text classifier for iOS cellular error log reports. Improved F1 score by 0.35 and deployed for online inference. Semi-automated data annotation of NL descriptions of cellular errors in Apple's iOS platform.

Goldman Sachs

Summer Analyst

Bangalore, India

Jun 2022 – Jul 2022

- Designed new database to store mathematical constraints for linear optimization models, reducing fetch times by 85%.

RESEARCH/PROJECTS

PyTorch Clone (needle) | CMU, Deep Learning Systems

(Python, C++, CUDA, Metal)

- Neural Network library with CUDA, Apple Silicon, and CPU backends, supporting standard initializers, modules (linear, conv, TransformerLayer), and autograd. Achieved matmul speedup of 80x with Apple Silicon and 52x speedup with CUDA (T4).

Efficient Pre-training of Language Models | Capstone Project, Advisor: Emma Strubell

(HuggingFace, PyTorch)

- Formulating "lazy pre-training", a method to grow model parameters from a smaller size to a larger size in a function-preserving manner and then continuing pre-training with fewer parameters and less data/compute.

Distributed Analytics on the Cloud | CMU, Cloud Computing

(Java, Spark, ETL, AWS, Databricks)

- Developed 3 distributed, efficient microservices: blockchain mining & verification, QR code, and Twitter user rec system.
- Performed large-scale ETL to reduce 1TB of raw data to 35GB. Serves 230,000 requests per second within budget of \$1.28/hr.

Interpreting Mathematical Reasoning | CMU, Advanced NLP

(HuggingFace, PyTorch)

- Using sparse autoencoders to study mathematical reasoning abilities in LLMs.

On-Device Image Captioning | CMU, On-Device Machine Learning

(CoreML, HuggingFace, PyTorch, Swift)

- iOS app that uploads a picture from camera roll and runs an int8 quantized image captioning model (BLIP) on-device.

Evaluation of Image Generation Models | Independent Study, Advisor: Yonatan Bisk

(HuggingFace, PyTorch)

- Find intrinsic representations of semantic meaning to evaluate understanding capabilities of diffusion-based image-generation models. Evaluating models' understanding of concepts such as color, size, object type etc.

SELECT PUBLICATIONS

- [1] ScriptONES: Sentiment-Conditioned Music Generation for Movie Scripts, Machine Learning for Audio Workshop, **NeurIPS 2023**
- [2] C3PO: A lightweight copying mechanism for translating pseudocode to code, Student Research Workshop, **AACL-IJCNLP 2022**
- [3] Multivariate Covid-19 forecasting with vaccinations as a factor: The case of India and USA, IEEE TENSYP 2022 and Workshop on Healthcare AI and COVID-19, **ICML 2022** (Track 2)
- [4] GraphCoReg: Co-training for regression on temporal graphs, *Best Student Paper*, International Workshop on Mining and Learning with Graphs, **ECML-PKDD 2022**
- [5] A Comparative Study of Algorithms for Intelligent Traffic Signal Control, **ICMLAS 2021**

SKILLS

Programming Languages: Python, Java, SQL, C/C++, R, C#, Objective-C

Libraries: PyTorch, pandas, numpy, scikit-learn, HuggingFace, NLTK, Keras, TensorFlow, CoreML

Machine Learning: pre-training, efficiency, fine-tuning, post-training, reinforcement learning, interpretability, ML systems

Cloud: Azure, GCP, AWS, Git, Azure ML, Vertx, React, Regex, MongoDB, SQL, large-scale ETL